

Exercise 2: Aggregation with GROUPING SETS, ROLLUP, and CUBE

Aim

To perform multidimensional data analysis using SQL aggregation features: GROUPING SETS, ROLLUP, and CUBE.

Problem Understanding

Sales reports often require summaries at different levels, such as by region, by product category, and overall totals. SQL provides advanced grouping operators to generate these summaries efficiently.

Tables Used

- Customers
- Orders
- OrderDetails
- Products

Implementation

- Joined the Orders, OrderDetails, Customers, and Products tables.
- Used GROUPING SETS to generate totals by Region, Category, and both Region and Category.
- Used ROLLUP to generate subtotals for each region and a grand total.
- Used CUBE to generate all possible combinations of Region and Category totals.

Difference Between GROUPING SETS, ROLLUP, and CUBE

Feature	Description
GROUPING SETS	Generates only the grouping combinations specified in the query.
ROLLUP	Generates hierarchical subtotals and a grand total.
CUBE	Generates all possible combinations of the specified columns along with the grand total.

Sample Output

```
Output

Status : Successfully executed

Time:      Memory:
0.0000 secs 4.652 Mb

Your Output

East|Electronics|6
East|Furniture|4
North|Electronics|15
South|Electronics|2
South|Furniture|8
West|Electronics|7
West|Furniture|3
East||10
North||15
```

```
|Electronics|30
|Furniture|15
East|Electronics|6
East|Furniture|4
North|Electronics|15
South|Electronics|2
South|Furniture|8
West|Electronics|7
West|Furniture|3
East||10
```

Conclusion

This exercise demonstrates how advanced SQL aggregation techniques can be used to generate detailed sales summaries. GROUPING SETS provides customized groupings, ROLLUP generates hierarchical subtotals, and CUBE produces all possible aggregation combinations for multidimensional analysis.